

Missing Layer Inference (MLI): A Governance Framework for Structural Failure in Learning-Based Decision Systems

Author: Chen-En Hou
2026.01

Abstract

As large-scale neural networks and automated decision systems are increasingly deployed, the primary source of system failure is gradually shifting from model performance deficiencies to distortions in decision structures. In many cases, while model accuracy remains stable, real-world system performance begins to degrade — a gap that existing analytical and engineering tools are unable to explain. Missing Layer Inference (MLI) is proposed in this context.

I. Problem Definition

In real-world decision systems, neural networks (DNN / CNN / RNN / Transformer), even when supported by sufficient data and well-trained models, remain subject to fundamental structural limitations: models can only learn patterns that are present in observable data and encoded within their structural assumptions, and therefore cannot learn factors that do not exist within either.

These unmodeled influencing factors — those not represented by the model's structure — constitute the Missing Layer.

Although they may not yet produce explicit prediction errors, they have already begun to affect:

- Decision efficiency
- System stability
- Risk accumulation

The purpose of Missing Layer Inference (MLI) is to establish an additional inference mechanism outside the neural network decision system that is specifically designed to identify such structural deficiencies. In other words, MLI does not replace neural networks; it functions as a complementary sensing and governance layer for neural network-based decision systems.

Applicable Scenarios

MLI is particularly relevant when:

- Neural networks have achieved acceptable performance, yet the system exhibits unexplained degradation or anomalies
- Decision consequences are critical and cannot rely solely on internal model confidence
- External influencing factors exist that have not yet been captured by available data or structural assumptions

Before conducting MLI, an adequacy screening must be performed to ensure that observed degradation or anomalies are not attributable to either of the following:

1. Insufficient Model Capability

- Inadequate architectural expressiveness
- Improper loss function design
- Evident underfitting or model mis-specification

2. Incorrect Problem Definition

- Mis-specified decision objectives
- Misalignment between evaluation metrics and real-world outcomes
- Contextual assumptions that deviate from actual operational conditions

Only after these issues have been excluded should the Missing Layer Inference process be initiated.

II. Core Issues of Missing Layer Inference (MLI)

Q1. Does the current neural network–based decision system contain critical influencing factors that are not captured by the model structure or the available data?

Q2. If so, what are these factors likely to be, and to what extent do they affect the efficiency and stability of the neural network’s outputs?

Q3. Before these factors are explicitly learned and validated by the model, should decision processes and resource allocations be proactively adjusted?

III. MLI Diagnostic Triggers

1. Structural Omission

Key Question:

Has the decision system overlooked critical influencing factors?

Typical manifestations include:

- Neural network performance on training and test datasets remains *apparently normal*
- Real-world system outcomes begin to deteriorate
- Conventional debugging and model diagnostics fail to identify the cause

2. Degradation Drift

Key Question:

Is system degradation propagating in a manner that cannot be explained by existing models?

Common indicators include:

- Declining operational efficiency
- Amplifying output volatility
- Escalating costs of correction and mitigation

3. Decision Blind Spot

Key Question:

Do important influencing factors exist that have not yet been recognized at the decision-making level, but are already affecting outcomes?

Typical signals include:

- Frontline operators perceive abnormal conditions
- Management reports indicate no apparent issues
- No major incident has yet occurred

IV. Definition of the MLI Pathway

Missing Layer Inference (MLI) is a continuous, multi-threshold, governance-based diagnostic and decision framework. Its purpose is to systematically eliminate existing explanatory sources within complex decision systems through layered screening, identify whether critical structural layers exist that have not yet been modeled but are already exerting substantive influence on system behavior, and guide subsequent interventions and resource allocation.

The MLI pathway is defined as follows.

Phase I — Problem Alignment and Accountability Clarification (Pre-MLI Screening)			
Step	Name	Functional Definition	Rationale
1	Adequacy Screening	Exclude causes originating from real-world misalignment, data bias, or institutional gray zones	Prevent misclassifying world-level problems as model problems
2	Model Alignment	Exclude causes arising from insufficient model capability or design flaws	Prevent misclassifying model defects as structural omissions
3	Problem Alignment	Exclude causes stemming from incorrect decision objectives or flawed problem formulation	Prevent searching for missing layers in the wrong problem space
Phase II — Missing Layer Diagnosis (MLI Detection)			
* Transforming decision systems that have excluded non-structural causes into a verifiable and governable set of Missing Layer hypotheses			
4	MLI Existence Assessment	Eliminate non-MLI factors through anomaly analysis, residual analysis, and counterfactual testing	Determine whether MLI activation is warranted
5	MLI Scope Delimitation	Derive potential types of missing structural layers and their impact ranges	Construct the candidate Missing Layer space
6	MLI Evidence Evaluation	Evaluate the strength and credibility of evidence for each candidate hypothesis	Control the risk of erroneous inference
7	MLI Prioritization	Rank hypotheses by impact magnitude and credibility	Focus decision-making resources
Phase III — Decision Intervention and Governance (Decision & Governance)			
8	Intervention Decision	Determine whether identified Missing Layers warrant decision-level intervention	Avoid excessive intervention
9	Intervention Modality Selection	Select technical, institutional, governance, or risk-based interventions	Align interventions with problem characteristics
10	Intervention Risk Management	Allocate resources and define accountability and risk ownership	Prevent accountability vacuums
11	Outcome Feedback	Track intervention outcomes and feed results back into the system	Establish a governance feedback loop

V. Mechanism and Process

Phase I — Problem Alignment and Accountability Clarification (Pre-MLI Screening)

Step 1. Adequacy Gate

All screening steps must be completed before MLI may be activated.

If the criteria are not satisfied, the use of this framework is not recommended.

1.1 Observational Alignment

Do the data currently used to train and operate the neural network truly represent the real-world phenomena they are assumed to represent?

1.2 Proxy Validity

Are critical real-world phenomena being represented by inappropriate, distorted, or progressively deteriorating proxy variables?

Typical Issues

- Are easily obtainable variables being used as substitutes for truly important variables?
- Does a misalignment exist where what is measurable is not what is important?
- Is the relationship between proxy variables and real-world phenomena degrading over time?
- Are metrics or proxies being manipulated or distorted due to institutional incentives or behavioral responses?

Diagnostic Signals

- Model performance remains stable while decision outcomes deteriorate
- Different data sources produce conflicting signals for the same phenomenon
- The gap between user perception and model outputs continues to widen
- Metrics improve while real-world problems remain unresolved

1.3 Observational Coverage

Are critical real-world conditions insufficiently captured by available data, or are decisions being made based on an outdated representation of reality due to reporting delays, temporal misalignment, or data drift?

Typical Issues

- Do “data voids” exist within the observable decision space?
- Are low-frequency, high-impact events entirely absent from the available records?
- Are specific populations, scenarios, time periods, or operational conditions systematically underrepresented or missing from the data?
- Do reporting delays, timestamp inconsistencies, or cross-system synchronization failures exist?

Diagnostic Signals

- The model persistently fails in specific scenarios.
- Errors are disproportionately large in edge cases.
- Decision correction costs are concentrated in under-observed or unobserved regions.
- System responses consistently lag behind changes in frontline conditions.

1.4 Institutional Unobservability

Do factors exist that are inherently unobservable due to institutional arrangements, incentive structures, power dynamics, or human behavior?

Typical Issues

- Are there behaviors that deliberately evade monitoring or auditing mechanisms?
- Do informal systems, shadow processes, or gray-market transactions exist?
- Do power structures or organizational culture obstruct information flow?
- Do incentive mechanisms inherently distort data generation?

Diagnostic Signals

- Significant discrepancies between inspection results and post-incident investigations
- Persistent mismatch between compliance data and real-world conditions
- Violations or incidents consistently originating from areas that appear “safest” in the data
- Long-standing problems that cannot be localized within available datasets

1.5 Observational Adequacy — Evaluation Criteria

Category	Sub-item	Indicator / Question	Type	Assessment Method	Outcome
Proxy Validity	Proxy Drift	Does the correlation between the proxy and target variables change significantly over time?	Quantitative	Δ corr, Stability Index	Pass / Flag
	Temporal Proxy Drift	Does the relationship between the proxy and real-world phenomenon degrade over time?	Quantitative	Δ corr over time	Pass / Flag
	Proxy Tension	Do model metrics diverge from actual outcomes?	Quantitative	KPI vs. Outcome Gap	Pass / Flag
	Proxy Redundancy	Do better proxy variables exist but remain unused?	Quantitative	Feature Attribution	Pass / Flag
	Adversarial Proxy Shift	Are indicators being manipulated due to institutional or behavioral responses?	Institutional	Expert Review / Field Investigation	Pass / Flag
	Proxy Validity	Do current proxies truly represent critical real-world factors?	Institutional	Expert Review	Pass / Flag
Observational Coverage	Coverage Gap	Do voids exist within the feature space?	Quantitative	Density Analysis / OOD Detection	Pass / Flag
	Rare–Impact	Are low-frequency, high-impact events systematically ignored?	Quantitative	Risk $\times$ Frequency	Pass / Flag
	Error Localization	Are errors concentrated in specific populations or scenarios?	Quantitative	Error Heatmap	Pass / Flag
	Observability	Is the extent of unobserved regions acceptable?	Institutional	Governance Assessment	Pass / Flag
Institutional Unobservability	Incentive Distortion	Do incentive structures encourage falsification or evasion?	Institutional	Institutional Review	Pass / Flag
	Shadow Process	Do gray processes or shadow mechanisms exist?	Institutional	Field Investigation	Pass / Flag

	Power Blocking	Do power structures obstruct information flow?	Institutional	Organizational Diagnostics	Pass / Flag
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Step 2. Model Adequacy

2.1 Model Capability Adequacy

Are the current model architecture and training coverage sufficient for the real-world decision contexts and problem structures?

Typical Issues

- Does the model exhibit evident underfitting or systematic errors?
- Is the model unable to capture known and reasonable relational structures?
- Has the model received sufficient training in critical decision scenarios?
- Does increasing model capacity or changing architecture significantly improve performance?

Diagnostic Signals

- Persistent failure even under controlled testing conditions
- Substantial performance improvement after increasing model capacity
- Structurally biased error distributions
- Significantly elevated error rates in specific scenarios or populations

2.2 Objective and Temporal Alignment

Are the objectives and time horizons optimized by the model aligned with actual decision requirements and environmental dynamics?

Typical Issues

- Is the loss function misaligned with real decision risks or cost structures?
- Do evaluation metrics obscure critical risks or long-term consequences?
- Is the model gradually aging due to environmental changes?
- Does the model update frequency lag behind real-world change dynamics?

Diagnostic Signals

- Continuous metric improvement without corresponding outcome improvement
- Unstable decision quality in high-risk scenarios
- Accelerating model failure in novel environments
- Clear degradation of decision effectiveness over time

2.3 Decision Stability and Adversarial Resilience

Can the model maintain reliable and stable decision behavior under uncertainty, perturbation, and adversarial conditions?

Typical Issues

- Do minor input variations lead to disproportionate output changes?
- Are certain populations or scenarios highly unstable?
- Do deliberate adversarial strategies exist that influence model outputs?
- Is the model susceptible to manipulation through rules, metrics, or institutional design?

Diagnostic Signals

- Similar cases yield significantly divergent predictions
- Inconsistent performance across scenarios
- Unpredictable oscillations in decision behavior
- High sensitivity of system performance to adversarial manipulation

2.4 Model Adequacy — Evaluation Criteria

Category	Sub-item	Indicator / Question	Type	Assessment Method	Outcome
Model Capability Adequacy	Architectural Expressiveness	Can the model fit known relationships?	Quantitative	Error Analysis / Fit Testing	Pass / Flag
	Training Coverage	Are decision scenarios adequately covered?	Quantitative	Coverage Analysis / OOD Detection	Pass / Flag
Objective & Temporal Alignment	Model Aging	Does performance degrade over time?	Quantitative	Performance Drift Analysis	Pass / Flag
	Objective Alignment	Is the loss function aligned with decision objectives?	Institutional	Expert Review	Pass / Flag
Decision Stability & Adversarial Resilience	Decision Stability	Is behavior stable under perturbation?	Quantitative	Sensitivity Testing	Pass / Flag
	Adversarial Risk	Is the model vulnerable to manipulation?	Institutional	Adversarial Testing	Pass / Flag
Decision Impact	High-Risk Stability	Is behavior stable in critical scenarios?	Quantitative	Stress Testing	Pass / Flag
	Decision Sensitivity	Do minor errors lead to major consequences?	Institutional	Risk Assessment	Pass / Flag
Controllability	Intervenability	Can human operators intervene or adjust the system?	Institutional	Process Review	Pass / Flag
	Failure Isolation	Can failures be contained?	Institutional	Architecture Review	Pass / Flag
Accountability Baseline	Traceability	Are decision sources traceable?	Institutional	Audit	Pass / Flag
	Auditability	Are decisions subject to post hoc review?	Institutional	Audit	Pass / Flag

Step 3. Problem Adequacy

3.1 Decision Framing Alignment

Is the currently modeled “problem” aligned with the actual decision task, and are decision boundaries, roles, and feasible actions clearly defined?

Typical Issues

- Is the system addressing a *prediction problem* or a *decision problem* — and are these two being conflated?
- Are the decision subject, decision authority, and decision frequency clearly specified (who decides, when, and on what)?
- Are decision boundaries correctly defined (what is outside the system’s responsibility)?
- Are multiple objectives being oversimplified into a single objective (e.g., focusing on efficiency while ignoring stability and risk)?
- Do the policy instruments and intervention mechanisms available to decision-makers correspond to the form of the model’s outputs?

Diagnostic Signals

- System predictions appear reasonable, yet provide no actionable guidance — or worse, lead to inappropriate actions
- Different stakeholders hold conflicting interpretations of what the system is actually solving
- The same output is interpreted differently by different departments, resulting in offsetting policy actions
- Decision disputes concentrate on *how the problem is defined* rather than *how accurate the model is*

3.2 Outcome and Label Validity

Do the outcomes, labels, or KPIs used by the model genuinely represent the results and risks that the decision process seeks to control?

Typical Issues

- Are the labels selected merely because they are *available* rather than *important* (KPI replacing purpose)?
- Do the indicators exhibit Goodhart’s Law effects, whereby metrics lose meaning once they become targets?
- Do the indicators ignore tail risks (low-probability, high-impact events) or long-term consequences?
- Are labels distorted by administrative procedures or reporting rules (e.g., under-reporting, delayed reporting, selective reporting)?
- Are outcomes that require multiple indicators being crudely approximated by a single metric?

Diagnostic Signals

- KPIs improve while actual risk events or external harm do not decline
- The organization begins “working for the metrics” while frontline conditions remain unimproved
- Indicators provide no warning or delayed response under high-risk conditions
- Persistent conflicts among indicators reveal that the objective function has not been properly constructed

3.3 Intervention and Feedback Loop Adequacy

Do decision actions, system feedback, and accountability form an operational closed loop that enables system correction rather than continuous drift?

Typical Issues

- Can system outputs be mapped to executable interventions (policy instruments,

inspection strategies, resource allocation)?

- Does a clear “intervention–observation–correction” loop exist, or does the system merely generate reports?
- Do decision-induced behavioral changes feed back into the data-generating process (policy feedback), yet remain unmodeled?
- Is outcome collection sufficient to determine whether interventions are effective (i.e., is there a genuine feedback signal)?
- Are decision authority and accountability aligned (those with authority bear responsibility; those responsible have authority)?

Diagnostic Signals

- After deployment, the system can only be observed but not adjusted — or adjustments cannot be validated
- The same problems recur and are repeatedly handled through ad hoc measures, preventing institutional learning
- After interventions, data distributions shift rapidly, rendering model behavior increasingly opaque
- Persistent misalignment between responsibility and authority reduces risk management to formality

3.4 Uncertainty Management Adequacy

Do decision and analytical processes adequately represent, manage, and absorb uncertainty, allowing controlled and reversible actions even when evidence is incomplete?

Typical Issues

- Are uncertainty ranges and confidence intervals explicitly communicated in model and analytical outputs?
- Do decision processes allow for deferred decisions, staged interventions, or low-consequence exploratory actions?
- Does institutional pressure exist to forcibly convert uncertainty into definitive directives?
- Is “not knowing” treated as an acceptable state rather than a failure?

Diagnostic Signals

- Decision documents and briefings present only single conclusions and single-point estimates
- Decision-makers later discover that substantial uncertainty existed but was never communicated
- High-risk decisions are prematurely finalized without buffering mechanisms
- Uncertainty is only discussed after decisions fail

3.5 Responsibility and Authority Alignment

Do decision authority, responsibility assignment, and risk consequences form a coherent and consistent mapping, allowing effective system correction when failures occur?

Typical Issues

- Does a structural pattern exist in which those with authority bear no responsibility and those responsible lack authority?
- Are risk consequences clearly attributable to specific decision units?
- Are critical decisions institutionally deflected to ambiguous collectives?
- Do responsibility shifting and risk offloading mechanisms exist?

Diagnostic Signals

- Responsibility cannot be localized after incidents
- Decision documents assign accountability ambiguously

- High-risk decisions frequently employ consensus language to diffuse responsibility
- Organizational culture systematically avoids named decision ownership

3.6 Problem Adequacy — Evaluation Criteria

Category	Sub-item	Indicator / Question	Type	Assessment Method	Outcome
Decision Structure Adequacy	Problem Definition	Does the problem correspond to the actual decision task?	Institutional	Expert Review	Pass / Flag
	Decision Boundary	Are decision boundaries clearly defined?	Institutional	Document Review	Pass / Flag
	Multi-objective Trade-offs	Are efficiency, risk, and stability objectives explicitly addressed?	Institutional	Decision Audit	Pass / Flag
Outcome & Label Adequacy	Label Validity	Do labels represent truly important outcomes?	Quantitative	KPI Cross-check	Pass / Flag
	Goodhart Risk	Are indicators susceptible to manipulation?	Institutional	Risk Assessment	Pass / Flag
	Tail Risk Coverage	Are low-probability, high-impact events covered?	Institutional	Scenario Testing	Pass / Flag
	Administrative Bias	Are labels distorted by administrative procedures?	Institutional	Process Audit	Pass / Flag
	Multi-metric Completeness	Should multiple indicators be used instead of a single proxy?	Institutional	Expert Review	Pass / Flag
Intervention & Feedback Adequacy	Actionability	Can outputs be translated into executable actions?	Institutional	Practical Validation	Pass / Flag
	Closed-loop Capability	Does an intervention–observation–correction loop exist?	Institutional	Process Review	Pass / Flag
	Policy Feedback	Are policy feedback effects incorporated?	Institutional	Expert Review	Pass / Flag
Uncertainty & Risk Preference Adequacy	Uncertainty Representation	Are uncertainty ranges and confidence intervals presented?	Institutional	Document Review	Pass / Flag
	Tiered Decision-Making	Are staged or exploratory interventions allowed?	Institutional	Process Review	Pass / Flag
	Risk Preference	Are decision-makers' risk preferences	Institutional	Organizationa	Pass / Flag

	Alignment	aligned?		1 Interviews	
Temporal & Accountability Boundary Adequacy	Temporal Alignment	Are decision, model, and policy time horizons aligned?	Institutional	Process Analysis	Pass / Flag
	Authority–Responsibility Alignment	Are authority and responsibility consistent?	Institutional	Organizational Audit	Pass / Flag
	Risk Offloading	Do responsibility shifting mechanisms exist?	Institutional	Organizational Diagnostics	Pass / Flag

Phase II — MLI Discovery Engine

MLI Detection Core

The MLI Detection Core transforms vague system degradation and decision anomalies into one or more concrete, testable **Missing Layer hypotheses**, enabling subsequent evidence evaluation and prioritization prior to entering the decision stage.

Process Stage	Description	Corresponding Steps
Inputs	Any system condition satisfying one or more of the following: • Performance degradation unexplained by existing models • Non-random distribution of system residuals • Frontline users perceive anomalies while management metrics remain normal • Persistent decision friction with rising costs	4
Anomaly Construction	Construct the Anomaly set A : • Explicitly define anomaly types • Annotate conditions under which anomalies occur • Describe anomaly evolution trajectories	5.2
Structural Slicing	Partition the decision space according to system structure to generate Slice set S , and observe anomaly behavior across structural segments	5.5 (a)
Structural Residual Test	Examine whether anomalies: • Can be explained by existing models, data, or decisions • Persist across most structural slices • Remain stable under counterfactual conditions Anomalies satisfying these criteria are classified as structural residuals	5.3–5.4
Counterfactual Stress Test	Intentionally remove or modify known factors and observe whether anomalies persist	5.5 (e)
VoI Gate	If VoI $\geq$ threshold , proceed to the Missing Layer hypothesis generation stage	5.5 (f)
Outputs	Generate candidate Missing Layer hypothesis set H , which serves as the formal input to Step 6 — MLI Evidence	6

	Evaluation	
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Step 4. MLI Existence Determination

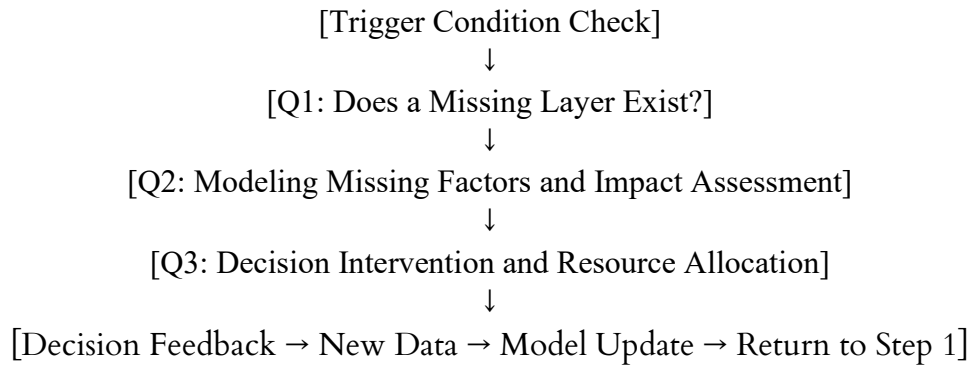
4.1 MLI Trigger Conditions

Activation

All evaluation items in **Steps 1–3** must pass (except for organizationally approved exception flags, for which the associated risks are explicitly assumed).

Prerequisite:

Decision Flow:



4.2 Threshold Gate

Proceed if **any two** of the following conditions are met:

Category	Indicator	Assessment Basis
Performance Gap	$\Delta\text{Perf} > \text{Threshold}$	Field performance vs. model prediction
Structural Residual	Non-random residuals	Residual clustering / patterned distribution
Decision Friction	Rising decision cost	Delays, conflicts, remedial actions
Behavioral Drift	Abnormal usage behavior	Operator behavioral shifts
Correction Failure	Repeated adjustments ineffective	Model tuning / institutional changes fail

Step 5. MLI Scope Identification

5.1 Q1: Determining the Existence of a Missing Layer (Structural Omission)

Purpose:

Step 5 is responsible for generating candidate hypotheses, not for determining their truth or value.

Method:

After Steps 1–3 have been passed (including accepted tolerance thresholds), if anomalies remain that are repeatable, structured, materially impactful, and not attributable to known causes, the existence of a Missing Layer is inferred.

All four conditions must be satisfied:

1. **Repeatable**
2. **Structured**
3. **Material**
4. **Not explained by known causes**

5.2 Anomaly Definition and Lock-in

Inputs

At least one from each category:

- Field performance / efficiency / stability indicators
- Decision cost indicators

- Model outputs or decision outputs

Outputs

A quantifiable anomaly set:

$$A=\{a1,a2,a3,...\}$$

Examples:

- Median delivery delay increases by 20% under identical input conditions
- Error rate of a specific population consistently exceeds the overall average by 2×
- Event handling cost (labor, time) rises while model metrics remain unchanged

If the **Anomaly set A cannot be explicitly defined**, Q1 fails and the process terminates.

5.3 Repeatability Test

A Missing Layer is not an isolated incident; it must be repeatable.

Indicators (at least one must pass)

- Anomaly persists across different time windows
- Anomaly persists across different samples or slices
- Anomaly persists across different regions or organizational units

Decision Criteria

- In at least **T consecutive observation windows**, **anomaly occurrence rate $\geq p$** (e.g., 60%)
- Or across **K slices**, at least **m** slices exhibit the same directional deviation

5.4 Structured Residual Test

The core of Q1 is not the presence of error, but the presence of **structured error**.

Methods (select at least two):

- **Residual Clustering:** Errors concentrate in specific scenarios, populations, or process nodes
- **Residual Patterning:** Errors exhibit persistent directional bias (consistently high or low)
- **Residual Association:** Residuals correlate strongly with recorded but unmodeled variables
- **Residual Migration:** Correcting one area causes errors to “move” to another — a classic structural omission signature

Decision Criterion

- The **random error hypothesis** is rejected (errors exhibit systematic structure)

5.5 Exclusion of Known Causes

Exclude three common illusion categories:

- **Data illusions** (related to Layer A)
- **Model illusions** (related to Layer B)
- **Problem framing illusions** (related to Layer C)

Full re-execution of Steps 1–3 is not required.

Q1 requires only **focused re-validation**.

Decision Criterion

- The anomaly slice passes the critical items in Steps 1–3 (or is flagged but within accepted tolerance)

Procedure

a. Lock anomaly slices

From **Anomaly set A**, extract:

- High-anomaly populations
- High-anomaly process nodes
- High-anomaly time windows

Obtain slice set:

$$S=\{s_1,s_2,\dots\}$$

b. Minimal necessary validation for each slice

For each s_i , perform the minimal validation checks in tables c and d below:

c. Local Re-validation Checklist

Group	Check Item	Diagnostic Question
A-Check	Proxy validity	Are the proxies used in this slice still appropriate?
	Observational coverage	Is data coverage missing or degraded in this slice?
	Institutional bias	Are there institutional or behavioral distortions affecting this slice?
B-Check	Model structure	Does this slice exhibit signs of underfitting?
	Training coverage	Is this slice outside the training distribution?
	Stability	Is model behavior highly unstable in this slice?
C-Check	Problem framing	Does this slice exceed the originally defined problem boundary?
	Metrics validity	Is this slice described by incorrect or misleading metrics?
	Intervention feasibility	Does this slice lack actionable interventions?

d. Local Re-validation Summary Table

Slice	A-Check	B-Check	C-Check	Conclusion
s_1	Pass	Pass	Pass	Candidate Missing Layer
s_2	Fail	—	—	Return to Layer A
s_3	Pass	Fail	—	Return to Layer B
...	...	...	...	...

Decision Rule

If **at least one slice** simultaneously satisfies:

- A-Check = Pass
- B-Check = Pass
- C-Check = Pass

and **structural anomalies persist**, then **Q1 is highly likely to hold**, and the process proceeds to the next verification stage.

e. Counterfactual Stress Test

For each anomalous slice s_i , conduct **at least one** counterfactual test:

Type	Intervention
Policy Counterfactual	Simulate alternative decision strategies
Process Counterfactual	Bypass specific process nodes
Resource Counterfactual	Modify resource allocation

Interpretation

- If anomalies remain stable across multiple counterfactuals → **Missing Layer hypothesis is strengthened**
- If anomalies disappear under counterfactuals → **Issue is decision-use error, not Missing Layer**

f. Value-of-Information (VoI) Gate

$$VoI = (DIP \times PR \times U) - IC$$

First term ($DIP \times PR \times U$): **value of acting**
 Second term (IC): **cost of exploring**

VoI Score	Action
≥ 40	Enter Q2 — hypothesis generation, 5.6–5.7 (high priority)
25–39	Enter Q2 — hypothesis generation, 5.6–5.7 (normal priority)
10–24	Low-cost exploration
< 10	Terminate MLI, return to monitoring

VoI Core Scale Definitions

Metric	Code	Score	Definition
Decision Improvement Potential	DIP	5	Prevents systemic failure or major accidents
		4	Significantly reduces risk or cost
		3	Improves efficiency or stability
		2	Minor optimization
		1	Negligible impact
Probability of Occurrence	PR	5	Repeatedly observed
		4	Appears across many contexts
		3	Appears in some contexts
		2	Occasional
		1	Rare
Exploration Cost	IC	5	Large-scale system redesign
		4	Long-term commitment required
		3	Cross-department coordination required
		2	Low-cost experimentation
		1	Negligible cost
Urgency	U	5	Immediate risk
		4	Near-term impact
		3	Mid-term impact
		2	Long-term impact
		1	No clear time pressure

5.6 MLI Inference Engine Overview

Upon completion of **Step 5 — Missing Layer Scope Identification**, the system formally enters the core MLI inference stage.

Purpose

To transform the following elements derived from Step 5:

- **Anomaly Set A**
 - **Structural Slices S**
 - **Structural Residuals R**
 - **Counterfactual Test Results C**
- into a set of **evaluable, rankable, and decision-ready Missing Layer hypotheses**

$$\{h_1, h_2, \dots, h_n\}$$

Prerequisites

- Q1 fully satisfied
- $VoI \geq 10$
- Complete availability of:

$$\{A, S, R, C, VoI\}$$

MLI Inference Pipeline

Anomaly $\rightarrow$ Structural Hypothesis Mapping



Missing Layer Impact Assessment



Evidence and Credibility Evaluation



Prioritization and Filtering



Gate: Decision Intervention Entry

5.7 MLI Cognitive Generation Mechanism — Structural Case Toolkit

This module constructs concrete **Missing Layer hypotheses** h_i from system evidence.

Conceptual Inference Relation

$$\{A, S, R, C, VoI\} \Rightarrow h_i$$

where h_i represents a **testable, falsifiable, and governable Missing Layer description**.

MLI Structural Case Toolkit (Core Template)

Component	Description	Case Entry
Anomaly Set A	Quantifiable abnormal phenomena	
Structural Slices S	Structural regions where anomalies concentrate	
Structural Residuals R	Behaviors unexplained by existing models	
Counterfactual Tests C	Persistence of anomalies after known factors are removed	
VoI	Value of investigating this issue	
Missing Layer Description (h_i)	Omitted structural factor	
Impact Mechanism	How the layer influences system behavior	
Impact Scope	Affected modules / processes	
Observable Prediction	Expected change if the layer is introduced	
Falsification Conditions	Evidence that would invalidate this hypothesis	

Illustration of MLI Outputs in Real-World Scenarios

This section illustrates the concrete outputs of **Missing Layer Inference (MLI)** in real-world contexts, enabling practitioners to understand **what MLI produces after completing the Detection Core**.

Case 1 — Night-Shift Failure in Supply Chain Operations

Over the past six months, a supply-chain system experienced an **18% decline in overall delivery efficiency**, while the model's prediction error remained within $\pm 2\%$.

Conventional debugging and process audits failed to identify the cause. Detection Core results for this case are summarized in the cross-case analysis below.

Case 2 — Emergency Department Triage Degradation

At a healthcare institution, patient boarding time in the emergency department **increased by 21% during holidays and night shifts**, while the patient-volume prediction model maintained accuracy above 95%. Management dashboards appeared normal, yet frontline medical staff consistently reported that “the system is becoming increasingly congested.”

Detection Core results for this case are summarized in the cross-case analysis below.

Case 3 — Distortion in Government Startup Subsidy Evaluation

A government startup subsidy program received **high performance predictions** from its evaluation model for three consecutive years, yet **actual industrial impact failed to materialize**, and post-hoc correction costs increased annually.

Detection Core results for this case are summarized in the cross-case analysis below.

Cross-Case Analysis

Component	Case 1	Case 2	Case 3
Anomaly Set A	Order delay +18%, return rate +9%	ED boarding +21%	Industrial impact unobserved
Structural Slices S	Night-shift logistics, third-party drivers	Night shifts, holidays	Startup firms
Structural Residuals R	Not explained by any of four model versions	Model predictions remain normal	Evaluation model scores remain high
Counterfactual Tests C	Anomaly persists after removing weather & order volume	Persists after removing patient flow	Persists after excluding funding factors
VoI	14.6	11.2	12.8
Missing Layer Description	Night-shift driver dispatch decisions unmodeled	Actual triage decisions deviate from design assumptions	Founders’ execution capability unmodeled
Impact Mechanism	Manual dispatch creates accumulated bottlenecks	Misclassification of patient severity	Execution gap
Impact Scope	Delivery efficiency, customer complaints	Bed utilization	Program effectiveness
Observable Prediction	Delay decreases after inclusion	Triage accuracy improves	Prediction improves with capability indicators
Falsification Conditions	No metric improvement after	No improvement in any indicators	No significant difference

	inclusion		
--	-----------	--	--

MLI Inference Results

Hypothesis h_1

“The actual dispatch decisions of third-party night-shift drivers are not represented in the current supply-chain model and have produced significant structural impact on system performance.”

→ Enter **Step 6: Evidence Evaluation**

Hypothesis h_2

“Actual triage decisions made by night-shift medical staff deviate substantially from system design assumptions, and this deviation is not captured by any model or process.”

→ Enter **Step 6: Evidence Evaluation**

Hypothesis h_3

“The real execution capability of founders and core teams is not incorporated into the subsidy evaluation model and constitutes a critical missing layer driving outcome distortion.”

→ Enter **Step 6: Evidence Evaluation**

Step 6. MLI Evidence Evaluation

6.1 Purpose and Role

After **Step 5** has established the high likelihood of a Missing Layer and passed the **Vol Gate**, the purpose of **Step 6** is to transform candidate Missing Layer hypotheses from *plausible explanations* into an *evidence base that decision-makers can stand behind*. This step controls inference risk and prevents organizations from undertaking high-cost interventions based on incorrect structural assumptions.

Step 6 does **not** answer whether MLI exists.

It answers whether a specific MLI hypothesis is supported by **sufficient evidence and credibility** to enter the decision layer.

6.2 Evidence Types and Source Definition

To avoid misjudgment driven by single-source observation, Step 6 requires coverage of at least the following evidence types:

Evidence Type	Description	Evaluation Criteria
Structural Evidence	Logical gaps in system architecture, processes, or institutional design	Explicit unmodeled structure (Yes/No)
Behavioral Evidence	Abnormal behavior of users, frontline staff, management	Cross-role consistency (High/Medium/Low)
Statistical Evidence	Residuals, error distributions, degradation curves	Residual deviation ($\geq 2\sigma$ = Strong)
Counterfactual Evidence	Whether the phenomenon disappears if the hypothesis holds	Anomaly disappears under hypothesis (Yes/No)
External Evidence	External cases, cross-system comparison, expert opinion	External case alignment (High/Medium/Low)

6.3 Evidence Strength and Credibility Scoring

For each candidate Missing Layer hypothesis h_i , quantitative scoring is performed. Each hypothesis must be evaluated against the following core principles:

- Whether it explains **multiple evidence sources simultaneously**
- Whether it is **compatible with known system mechanisms**
- Whether **reasonable alternative explanations** have been excluded

Any hypothesis that explains only a single evidence source is classified as **low credibility**.

The three core principles above are operationalized as follows:

Evaluation Item	0 Points (Insufficient)	1 Point (Moderate)	2 Points (Strong)
Evidence Volume	Only one source; not firsthand or lacks original records	≥2 independent sources, at least one indirect	≥2 independent firsthand sources, at least one with auditable original records
Evidence Consistency	Evidence conflicts; no coherent explanation	Most evidence supports; unresolved conflicts remain	Primary evidence highly consistent; no material conflicts
Explanatory Power	Explains only isolated or local anomalies	Explains multiple anomalies but leaves major gaps	Explains the core anomaly set and its evolution
Alternative Exclusion	No systematic exclusion of major alternatives	Partial exclusion of major alternatives	All known major alternatives excluded with documentation
Verifiability	No feasible verification	Verifiable but with prohibitive cost, risk, or time	Verifiable within acceptable cost, risk, and time
(B) Spurious Correlation & Alternative Explanation Exclusion Checklist (Penalty Mechanism)			
Short-term incidental phenomenon		☐ Yes ☐ No	
External shock event		☐ Yes ☐ No	
Data bias		☐ Yes ☐ No	
Model defect		☐ Yes ☐ No	
• Each "Yes" → Total score -1 • ≥3 "Yes" → Hypothesis marked Deferred • All 4 "Yes" → Hypothesis Excluded			
Total Score Interpretation (0~10)			
8–10	High credibility	Proceed to Step 7	
5–7	Medium credibility	Collect additional evidence and reassess	
0–4	Low credibility	Terminate	
Excluded	-	→ Archive	

Alternative Explanation Exclusion Requirements

At least **two** of the following must be completed and documented:

- Cross-validation against known model defects
- Cross-validation against data bias hypotheses
- Cross-validation against external shock events
- Cross-validation against known structural factors

No documentation → **2 points not allowed**

Verifiability Requirements

All of the following must be satisfied:

- Testable design available (experiment, A/B test, backtesting, simulation, sandbox)
- Completed within decision-acceptable timeframe
- **Cost $\leq$ 30% of potential risk impact**
- Verification will not cause irreversible damage to core system

Otherwise, the maximum score allowed is **1 point**.

Step 7. MLI Prioritization

Purpose:

To rank multiple high-credibility and medium-credibility MLI hypotheses according to their systemic impact risk and remediation cost, ensuring that limited decision resources are concentrated on the highest governance-value targets.

Role:

The final filtering and ordering mechanism between the MLI engine and the decision layer.

MLI Prioritization Decision Table

Hypothesis	Impact Severity (I)	Probability (P)	Diffusion (D)	Treatability (T)	Remediation Cost (C)	Priority Index = $(I \times P \times D \times T) \div C$
h_1	1–5	1–5	1–5	1–5	1–5	
h_2	1–5	1–5	1–5	1–5	1–5	
...	...	...	...	...	...	...

Priority levels: ≥ 60 = Critical | 30–59 = High | 10–29 = Medium | < 10 = Low

Scoring Criteria

Score	I — Impact Severity	P — Probability	D — Diffusion	T — Treatability	C — Remediation Cost
1	Single-metric impact	$<1\%$ per year (period)	Single node	Nearly non-intervenable	<1 person-week
2	Local performance impact	1–5% per year (period)	Adjacent processes	High difficulty	1–4 person-weeks
3	Cross-process impact	5–20% per year (period)	Multiple processes	Intervenable with high cost	1–3 person-months
4	Systemic / regulatory impact	20–50% per year (period)	Cross-system	Feasible	3–12 person-months
5	System failure / major incident	$>50\%$ per year (period)	Cascading failure	Easily and rapidly intervenable	>12 person-months / major political cost

Phase III — MLI Decision and Governance

Phase III marks the formal transition of the MLI process from analysis and diagnosis into the domain of decision and governance.

At this stage, a Missing Layer is no longer merely a systemic hypothesis; it becomes a governance issue for which the organization must assume responsibility, allocate resources, and bear risk.

Mission of This Phase

To transform the Missing Layers identified and prioritized in Phase II into **executable decision interventions**, while establishing comprehensive accountability structures, risk controls, and feedback mechanisms to ensure system stability and sustainable governance.

Key Governance Questions

- Should we intervene?
- How should we intervene?
- Who is responsible?
- What risks are being assumed?
- How will outcomes be evaluated and corrective actions determined?

Step 8. Intervention Decision Gate

Intervention Decision Matrix

Dimension	Evaluation Question	Yes	No
Evidence Maturity	Is Step 6 credibility rated High?	☐	☐
Governance Priority	Is Step 7 priority High or Critical?	☐	☐
Risk Necessity	Is the risk unacceptable if no intervention occurs?	☐	☐
Solution Feasibility	Are executable intervention options available?	☐	☐
Accountability Capacity	Are responsible entities and resources clearly assigned?	☐	☐
Decision Rule • All Yes → Proceed to Step 9 • Any No → Intervention prohibited; register as a governance monitoring item and reassess periodically			

Step 9. Intervention Mode Selection

Objective:

To translate each Missing Layer hypothesis h_i into a concrete intervention plan — specifying **what to modify, what data to add, what modules to introduce, which processes to adjust, and which institutional rules to revise** — and to select the **Minimum Viable Intervention (MVI)**.

Case 1 — Supply Chain Night-Shift Failure

(Unmodeled Dispatch Decisions of Third-Party Night-Shift Drivers)

Known Inputs (from Step 5.7):

- **A**: Delivery delay +18%, return rate +9%, dispatch mismatch +22%
- **S**: Night-shift logistics, third-party driver slices
- **Vol**: 14.6
- **h_I** : Actual dispatch decision behaviors of third-party night-shift drivers are not modeled and have structural impact

Gap Assessment:

- Missing real dispatch data (dispatch rules, overrides, late acceptance, no-shows, route deviations)
- Behavioral variability of third-party drivers (delayed acceptance, order refusal, last-minute shift changes, fatigue effects)
- Institutional differences in night-shift operations (night premiums, control policies, handover rules)

Intervention Targets:

Intervention Type	Intervention Decision	Deliverables
Data	Complete night-shift third-party driver event logs (accept/refuse/cancel/timeout/reroute/handover)	Event table schema + ETL + daily completeness report
Model	Introduce “driver availability/reliability” features into scheduling/prediction	Feature definitions + baseline experiments
Process	Require override logging & backfill in night-shift dispatch	Override fields + mandatory rules
Governance	Align third-party KPIs & contracts with night-shift risks (refusal rate, no-show)	KPI clauses draft + monitoring dashboard

Minimum Viable Intervention (MVI):

MVI-1: Data + Process intervention first

- Build night-shift driver event logs + mandatory override recording
- Goal: Make the Missing Layer observable before attempting full correction

Step 9 Completion Criteria:

- Night-shift driver event table completeness $\geq 90\%$
- Override reason coverage $\geq 80\%$
- Anomalies in **A** decomposed into 2–3 reproducible patterns under slice **S**

Case 2 — Emergency Department Triage Degradation

(Unmodeled Night-Shift Triage Decision Bias)

Known Inputs:

- **A:** ED boarding +21%, bed allocation delay +17%
- **S:** Night shifts, holidays
- **VoI:** 11.2
- ***h*₂:** Night-shift triage behavior deviates from system assumptions and is unmodeled

Gap Assessment:

- Actual triage rule usage (how, when, why adjustments occur)
- Night-shift resource constraints (staff, beds, diagnostics waiting time)
- Implicit objectives (reducing boarding vs avoiding complaints)

Intervention Targets:

Intervention Type	Intervention Decision	Deliverables
Data	Introduce triage reason codes + revision trail	Triage log schema + revision markers
Process	Night-shift triggered secondary review mechanism	Review thresholds + SOP
Model	Add night-shift constraint features to prediction/scheduling	Constraint features + benchmark
Governance	Shift night-shift KPIs from speed to correctness × safety	KPI revision + exception handling

Minimum Viable Intervention (MVI):

MVI-2: Build triage audit trail + triggered secondary review

- Do not immediately rebuild the triage model
- First make the deviation measurable and traceable

Step 9 Completion Criteria:

- Night-shift triage reason coverage $\geq 85\%$
- Revision traceability (who/when/why) $\geq 80\%$
- At least one actionable process point identified for ED congestion anomaly

Case 3 — Subsidy Evaluation Distortion

(Unmodeled Team Execution Capability)

Known Inputs:

- **A:** High evaluation scores with poor outcomes; rising correction cost
- **S:** Startups, early-stage teams
- **VoI:** 12.8
- ***h*₃:** Founder/core team execution capability missing from evaluation model

Gap Assessment:

- Missing execution proxies (delivery history, milestone punctuality, team stability)
- Incentive distortion (writing quality mistaken for execution ability)
- Insufficient governance gates for early-stage teams

Intervention Targets:

Intervention Type	Intervention Decision	Deliverables
Data	Build milestone punctuality/change/delivery quality history	Milestone table + data dictionary
Process	Implement staged funding + gate reviews	Gate design +

		disbursement rules
Model	Introduce execution capability proxies	Proxy indicators + weighting scheme
Governance	Define execution risk as veto condition	Veto rules + evidence requirements

Minimum Viable Intervention (MVI):

MVI-3: Staged funding gates + milestone datafication

- Make “ability to execute” traceable before modeling it
- Most aligned with government governance context and easiest to implement

Step 9 Completion Criteria:

- Milestone fields defined and consistently reported ($\geq 90\%$ coverage)
- Gate reviews institutionalized (e.g., Month 2 / Month 4 checkpoints)
- At least one data pattern distinguishes “high-score failure” from “high-score success”

Step 10. Intervention Risk and Accountability Layer

After the **Minimum Viable Intervention (MVI)** is selected in **Step 9**, this step translates the intervention plan into explicit definitions of **responsible parties, costs, risks, and failure-response strategies**.

P — Risk Probability (1–5)

Assessment horizon: the duration of the current intervention plan (typically 30–90 days)

Score	Level	Definition
1	Very Low	<5% probability; no historical occurrence; no known triggers
2	Low	5–15%; occasional; limited precedent
3	Medium	15–40%; occurred in similar scenarios
4	High	40–70%; frequent historical occurrence
5	Very High	>70%; near-certain occurrence

I — Risk Impact (1–5)

Score	Level	Definition
1	Minor	Negligible impact; no reporting required
2	Controllable	Local impact; recoverable within unit
3	Significant	Cross-department impact; management intervention required
4	Severe	Major operational / safety / reputational risk
5	Catastrophic	Regulatory, life, safety, or organizational survival risk

R — Risk Level = $P \times I$

R Range	Risk Level	Decision Rule
1–4	Low	Executable; unit-level control
5–9	Medium	Supervisor approval required
10–15	High	Senior management / committee approval required
16–25	Extreme	Execution prohibited unless approved by highest authority

Rollback Risk (1–5)

Difficulty and cost of reverting to original state after intervention failure

Score	Level	Definition
1	Minor	Immediate rollback; no additional loss
2	Controllable	Low cost; recovery within days
3	Significant	Moderate cost; recovery requires weeks
4	Severe	High cost; rollback itself introduces secondary risk
5	Catastrophic	Nearly irreversible or rollback itself causes disaster

Accountability Level (A–D)

Level	Responsibility Scope
A — Departmental	Manager accountable (performance / resource adjustments)
B — Cross-Departmental	Senior management / committee accountable (major resource / strategic changes)
C — Institutional	Chief responsible authority accountable (external, regulatory, major incidents)
D — Regulatory	Potential regulatory, medical, or public safety liability; prior approval mandatory

Intervention Risk & Accountability Matrix — Case Study

Case	Step 9 Selected MVI (Intervention Target)	Primary Risk Event (Top-1)	P	I	R = P × I	Risk Tolerance Gate (Quantitative)	Escalation Trigger	Rollback Design (Reversibility)	Rollback Risk	Resource Requirements (People / Cost / Timeline)
1. Supply Chain Night-Shift Drivers	Night-shift third-party driver event logs + mandatory override fields (observability first)	Third-party misreporting causing distorted data and misleading diagnosis	4	3	12 (High)	Within 60 days: event completeness ≥90%, override coverage ≥80%; operational KPI deterioration ≤5%	KPI deterioration >5% or completeness <80% for 2 consecutive weeks	Reversible: stop schema expansion, revert process; retain collected data but mark unreliable	3	People: Ops 1 + Data/IT 1; Cost: system modification + vendor coordination; Timeline: 6–8 weeks
2. ED Night-Shift Triage	Triage audit trail + triggered secondary review (pilot)	Slower process or increased mis-triage causing patient safety incident	3	5	15 (High)	Within 30 days: boarding time increase ≤10%; mis-triage not increased; reason code coverage ≥85%	Boarding >10% or mis-triage increase → immediate escalation; any major incident → immediate	Strongly reversible: disable secondary review, revert triage; retain audit trail	4	People: ED clinical 1 + IT 1; Cost: HIS modification; Timeline: 8–10 weeks (incl. training)

							suspension			
3. Subsidy Evaluation Distortion	Staged funding gate + milestone datafication (institutional first)	Increased administrative burden leading to execution stagnation / backlash	3	3	9 (Medium)	Administrative cost increase ≤20%; execution rate ≥70%; milestone coverage ≥90%	Admin cost >20% or execution rate <70% for 1 period	Reversible: narrow scope (specific categories / amounts), simplify gates; no full system rollback	2	People: PMO + reviewers; Cost: review operations; Timeline: 1 subsidy cycle (pilot)

Risk Scoring Formula & Threshold Reference

$$R=P \times I$$

Intervention Rule:

If **R ≤ 15** and **Rollback Risk ≤ 3**, intervention is permitted. Otherwise, escalation to higher accountability level is required, or intervention must be terminated.

Step 11. Outcome and Structural Learning

Purpose:

To transform the intervention results from **Steps 9–10** into new structural evidence, update the credibility, prioritization, and governance judgments of MLI, and thereby close the learning loop.

11.1 Feedback Core

$$\text{Intervention Value} = \Delta \text{Outcome} - \Delta \text{Risk} - \Delta \text{Cost}$$

Feedback Focus Areas:

- **Step 6:** Evidence strength
- **Step 7:** MLI prioritization
- **Step 8:** Future intervention decisions
- **Model / Problem / Decision structures**

11.2 Feedback Process and Evaluation Table

Steps 9–10 Intervention Execution



Collect Outcome / KPI / Incident / Feedback



Compute $\Delta \text{Outcome}$, ΔRisk , ΔCost



Update MLI Evidence and Credibility



Adjust Future Intervention Strategy and Structural Design

Performance Evaluation Table

Dimension	Metric	Calculation	Threshold	Feedback Action
Outcome Improvement ($\Delta \text{Outcome}$)	Core KPI improvement	(After – Before) / Before	$\geq +10\%$	Increase MLI evidence strength
Risk Change (ΔRisk)	Risk level shift	$R_{\text{before}} - R_{\text{after}}$	≥ 3	Confirm intervention effectiveness
Cost Change (ΔCost)	Resource consumption	Actual – Estimated cost	$\leq +10\%$	Within control
Incidents & Anomalies	Major events	Count	0	Maintain strategy
User Feedback	Frontline subjective improvement	Survey interview /	$\geq 70\%$ positive	Support hypothesis

11.3 Feedback Threshold Design

Condition	Action
$\Delta \text{Outcome} \geq 10\%$ and $\Delta \text{Risk} \geq 3$	Institutionalize the structural correction
$\Delta \text{Outcome} < 5\%$ or $\Delta \text{Risk} < 1$	Reduce MLI weight
Emergence of new risks	Return to Steps 5–6 for reassessment
Intervention failure	Trigger rollback and downgrade MLI

11.4 Step 11 Outputs

Focused deliverables:

- Updated Missing Layer hypothesis credibility

- Updated governance thresholds and intervention strategies
- Updated model and decision structure design principles
- Organizational knowledge base construction

VI. System Structure, Model Design & Usage

12.1 Architectural Placement

MLI as a post-decision, cross-layer meta-diagnostic layer

World → Data → Model → Decision → Action → Outcome

↓

MLI

↓

Model / Decision / Structure Revision

When to Use

- Accumulating anomalies
- Metric degradation
- Prior to high-impact decisions
- Early warning signals of system failure

Control Boundaries

All structural revisions, objective adjustments, and intervention decisions **must pass explicit accountability and governance decision gates.**

12.2 MLI Module Placement

The optimal insertion point of the MLI module is the **post-inference / post-decision observation layer.**

Required Standardized Observational Signals

- | | | |
|--------------------|--|-----------------|
| • Input | request metadata, key feature hash/embedding, slice fields | summary |
| • Model | prediction/score, confidence, optional logits | outputs |
| • Decisions | & | actions |
| | adopted/rejected actions, downstream policies, resource allocation | |
| • Outcome | delayed labels, true KPIs, incidents/anomalies, human overrides | feedback |
| • Residual | & | anomaly |
| | error, drift, volatility, rising costs, etc. | metrics |

12.3 Multi-Model Integration

a. Data Contract

Define a **minimal, universal event schema** (JSON / Avro / Protobuf recommended):

- **system_id, model_id, model_version**
- **timestamp**
- **slice_keys** (structural slices S in Steps 5–6)
- **input_fingerprint** (irreversible hash or embedding ID for privacy)
- **output** (score/label + confidence)
- **decision** (action ID, policy ID, override flag)
- **outcome** (delayed feedback: KPI, ground truth, incidents)
- **diagnostics** (drift metrics, latency, resource usage)

b. Adapters / Connectors

Each system implements a one-time adapter converting existing logs / DBs / message queues into the standard schema.

- **Online inference:** API gateway / inference server logs

- **Data:** feature store / data warehouse
- **Decisions:** policy engine / workflow system
- **Feedback:** monitoring systems, human labeling, customer service tickets, audit results

c. MLI Engine Core

The engine **only reads the standardized event stream** and outputs:

- **Detection:** anomaly set A, slices S, residuals R, counterfactual tests C, VoI gate
- **Evidence:** Step 6 scoring
- **Prioritization:** Step 7 ranking
- **Decision Gate:** Step 8 intervention decision
- **Plans:** Steps 9–11 intervention & feedback
(*primarily recommendations; execution remains human / process-controlled*)

12.4 Model Training Layers

Path 1 — Rule & Scale Calibration (fastest, most universal)

- Threshold calibration: VoI, residual anomaly thresholds, slice significance
- Evidence scoring calibration: operational definitions & mappings in Step 6
- Prioritization weight calibration: dimensions & scales in Step 7

Framework-agnostic, fully cross-model compatible.

Path 2 — Optional Dedicated MLI Assist Models

Examples:

- Anomaly → hypothesis recommendation engine (h_i generation assistant)
- Evidence consistency & explanatory power scoring assistant
- Intervention plan recommendation (*still subject to Step 8 gate*)

These models consume only the standardized event schema and **do not directly couple to underlying neural networks**.

Incremental Adoption Path

Initial inputs:

- **Online:** (input_fingerprint, output, decision)
- **Offline:** (slice_keys, KPI/outcome, incident)

After completing the Detection Core (A / S / R / C / VoI), expand into Steps 6–11.

12.5 Deployment Patterns

Sidecar Service

- Deployed alongside inference / decision systems
- Performs event collection & feedback aggregation
- Low intrusion, high organizational acceptance

Plugin SDK

Minimal SDKs for event generation and message publishing (Kafka / PubSub / HTTP)

- Python (ML workloads)
- Java / Go (backend services)
- JS / TS (front-end & workflow)

12.6 Compatibility Risks & Mitigations

1. Inconsistent Slice Fields (S)

Mitigation: configurable slice_keys dictionary in schema; engine computes both minimal common set and system-specific extensions.

2. Delayed Feedback & Sparse Labels (RWD / RWE common)

Mitigation: built-in delayed feedback windows (e.g., 7 / 30 / 90 days); proxy outcomes allowed until true labels arrive, then evidence is updated.

3. Data Privacy & Access Control

Mitigation: no raw inputs stored; fingerprints only; sensitive fields

hashed/bucketed; evidence chain remains auditable without exposing plaintext.

Acknowledgement

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